

Joy Sarker

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RESEARCH INTERESTS

Agentic AI and LLM tool use, especially how coding agents can check whether their own output is correct. Related interests: self-critique, learning from execution feedback, evaluating agents on large real-world codebases, and planning under limited compute.

EDUCATION

State University of Bangladesh

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

2018 – 2022

- CGPA: **3.96 / 4.00**; Batch Topper (highest CGPA in the batch)
- Dean's Award (Silver Medal), 6th Convocation, 2022; Merit Scholarship in every semester
- Relevant coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Algorithm Analysis, Theory of Computation, Discrete Mathematics, Software Engineering

PROFESSIONAL EXPERIENCE

Therap (BD) Ltd.

Dhaka, Bangladesh

Software Engineer II (Associate SE → SE → SE II)

June 2022 – Present

Bangladesh affiliate of Therap Services (US), which makes software for human services providers.

- Develop features with Spring MVC, JSP and JavaScript from business requirement documents.
- Write JUnit unit tests and Playwright end-to-end tests for every feature.
- Fix security vulnerabilities in the application and write tests that confirm each fix.
- Review code for internal and external teams.
- Moved critical business logic from the client to the server to improve security and data integrity.
- Redesigned the UI and backend of a core module to make it easier to maintain.
- Optimized database queries, reducing execution time by 30%.
- Mentored 5+ junior developers in Spring and JPA.

Stack: Java, Spring MVC, JSP, JPA/Hibernate, Oracle SQL, JavaScript, JUnit, Playwright, Gradle

ACADEMIC PROJECTS

Algorithm & Data Structure Visualization Engine [\[code\]](#) [\[demo\]](#)

2022

B.Sc. final-year project, with M. Huq and K. Somapriya

Web app that animates sorting algorithms (bubble, merge and quick sort) and graph algorithms (Dijkstra, BFS, DFS) step by step. Users can enter custom inputs to compare best-case and worst-case behavior.

Decision Tree and k-NN Classification (Machine Learning course project) [\[code\]](#) [\[report\]](#)

2021

Built Decision Tree and k-NN classifiers with scikit-learn on a real-world dataset. Compared $k = 1$ to 5 to check for overfitting and evaluated the models with confusion matrices.

HONORS & AWARDS

Dean's Award (Silver Medal), State University of Bangladesh

2022

Honorable Mention, ICPC Asia Dhaka Regional

[2019](#), [2020](#), [2021](#)

Top 8% worldwide, Facebook Hacker Cup (rank 2826 of 35,000+)

2021

Qualified for Round 1, Google Code Jam

2022

Codeforces Specialist ([PrimeX](#)); solved 2000+ problems on online judges

TEACHING & SERVICE

Competitive Programming Trainer, State University of Bangladesh Mar 2020 – Aug 2022
Trained junior students for programming contests.

Problem Setter and Judge, ICPC-style contests on Toph, Dept. of CSE, SUB 2021 – 2024
Wrote problems for four contests, including [SUB CSE Tech Carnival 2024](#) and [SUB CSE Fest IDPC 2023](#).

Invited Speaker, Code Quest Workshop on Competitive Programming, Dept. of CSE, SUB Aug 2023

President, SUB Computing Club Sep 2021 – Jun 2022
Organized programming contests, seminars and workshops for the CSE department.

SKILLS

Languages	Python, Java, C/C++, JavaScript
Frameworks	Spring MVC, Spring Boot, JPA/Hibernate, scikit-learn
Testing	JUnit, Playwright, security testing
Databases	Oracle SQL, MySQL
Tools	Git, Gradle, Docker